VANTRACK PALAWAN

Shuttle Van Terminal Schedule, Seat Availability, Booking and Rental Management System

A Flutter-Based Mobile Application Project

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INTRODUCTION

VanTrack Palawan is a mobile application designed to improve the organization and management of shuttle van transportation services in Palawan. The application aims to provide passengers with a convenient platform where they can search for available trips, check departure schedules, view seat availability, and manage their transportation bookings. It also aims to support van operators and authorized personnel in managing vehicle information, van rental requests, inventory records, and transportation-related transactions.

The proposed application will be developed using the Flutter framework, with Dart as its primary programming language. Flutter is selected as the main technology stack because it provides a modern approach to mobile application development through a single codebase that supports multiple platforms, particularly Android and iOS.

Flutter is an appropriate technology for VanTrack Palawan because of its cross-platform capabilities, extensive collection of customizable widgets, and strong developer community. These advantages allow the development of responsive user interfaces, reusable application components, and consistent functionality across supported devices.

Furthermore, Flutter provides development tools such as Hot Reload, which allows developers to quickly observe changes during implementation. Its documentation, packages, and community resources also provide valuable support throughout the development, testing, and maintenance of the application.

PROJECT OVERVIEW

VanTrack Palawan is designed as a digital transportation management platform that organizes shuttle van schedules, passenger bookings, seat availability, van rental requests, fleet information, inventory records, and transaction histories.

The project will utilize Flutter's cross-platform development capabilities to support both Android and iOS devices. By maintaining a shared codebase, the development process can reduce duplicated programming work, minimize development time, and potentially lower maintenance costs.

The application will take advantage of several important Flutter features:

A. Cross-Platform Development

Flutter allows the development of applications for Android and iOS using a shared codebase. This approach will help provide a consistent user experience across different mobile platforms.

B. Hot Reload

The Hot Reload feature will be used to speed up the development process by allowing developers to see many code changes immediately without restarting the entire application.

C. Widget-Based Architecture

Flutter's widget-based architecture will be utilized to create reusable interface components, including buttons, forms, navigation bars, dropdown menus, dashboard panels, and trip information cards.

D. Responsive User Interface

Responsive layouts will be implemented to help the application adapt to different screen sizes and device orientations, ensuring that the interface remains organized and accessible.

E. Material Design Components

Flutter's Material Design widgets will be used to create a professional and consistent interface for the application's navigation, forms, menus, and interactive elements.

Through these capabilities, VanTrack Palawan aims to provide an organized, accessible, and user-friendly transportation management experience.

TARGET AUDIENCE

The primary target audience of VanTrack Palawan consists of passengers, commuters, van operators, terminal personnel, and system administrators involved in shuttle van transportation services in Palawan.

Flutter's cross-platform compatibility will allow the application to serve users with Android and iOS devices, making the system accessible to a broader range of mobile users.

A. Passengers and Commuters

- Students traveling between municipalities and destinations in Palawan.
- Employees and working individuals who regularly use shuttle van transportation.

- Local residents who need convenient transportation services.

- Tourists and visitors who require transportation between destinations.

- Passengers who want to check trip schedules, available seats, fares, and booking information.

B. Van Operators and Terminal Staff

- Transportation operators responsible for managing shuttle van services.

- Terminal personnel who organize departure schedules and passenger bookings.

- Staff members who monitor vehicle availability and trip information.

- Authorized personnel who handle rental requests and transportation records.

C. System Administrators and Dispatchers

- Administrators responsible for managing system information.

- Dispatchers who monitor transportation schedules and vehicle assignments.

- Personnel responsible for maintaining fleet and inventory records.

- Authorized users who review transactions, activities, and operational reports.

The application is primarily intended for adult passengers and transportation personnel who use Android or iOS mobile devices. Its interface will focus on accessibility, clear navigation, and efficient access to transportation information.

FEATURES

VanTrack Palawan will utilize Flutter's widget-based architecture to develop its user interface and application functionality efficiently. Reusable widgets will be implemented to maintain consistency throughout the application, particularly in its forms, buttons, navigation components, trip information panels, and management dashboards.

The following are the major features of the proposed application.

A. User Account Management

- Sign In
- Create Account
- User Profile
- Account Information
- Account Management

B. Dashboard

- Professional dashboard interface
- Quick access to important application modules
- Upcoming trip information
- Notifications and updates
- Navigation to other system features

C. Trip Search and Schedule Management

- Select departure location
- Select destination
- Choose travel date
- Search available shuttle van trips
- View departure schedules
- Display trip fares and information

D. Trip Details and Seat Availability

- View van and trip information
- Display available seats
- Select preferred passenger seats
- Review trip information before booking
- Monitor seat availability

E. Passenger Booking Management

- Submit passenger booking requests
- Booking confirmation
- View existing bookings
- Booking history

- Modify or cancel booking requests

- Review booking status

F. Van Rental Management

- Complete van rental request forms

- Enter pick-up and drop-off locations

- Select rental dates and duration

- Review rental packages

- Submit rental requests

- Maintain rental records

G. Fleet and Vehicle Management

- Maintain vehicle information

- Monitor vehicle availability

- Record vehicle details

- Organize fleet information

- Manage vehicle records

H. Inventory Management

- Maintain operational supplies and resources

- Monitor inventory quantities

- Record inventory additions and releases
- Review inventory activity
- Maintain inventory records

I. Notifications and Customer Support

- Display important booking notifications
- Provide help and support information
- Communicate relevant trip updates
- Display booking and rental status updates

J. Transaction History and Reports

- Maintain booking and rental activity records
- Review completed transactions
- Display booking and rental summaries
- Monitor recorded activities
- Generate operational reports for authorized personnel

Flutter Packages and Plugins

The project will utilize Flutter's built-in widgets and development tools, together with additional packages when required.

The following technologies and packages are considered for the application:

Package / Technology Intended Purpose

Flutter SDK Main framework for application development

Dart Programming language

Material Design Widgets User interface and application components

Cupertino Icons Icon resources for interface elements

shared_preferences Local storage of simple application preferences

sqflite Potential local SQLite database storage

Flutter Test Automated testing of application components

Additional packages for authentication, notifications, and backend connectivity may be integrated depending on the application's implementation requirements.

FLUTTER PACKAGES AND TECHNOLOGIES

The development of VanTrack Palawan will utilize the Flutter ecosystem and supporting development tools to implement its user interface, application logic, and testing environment.

The following technologies will be used:

A. Flutter SDK

Flutter will serve as the primary framework for developing the application's interface and mobile application functionality.

B. Dart Programming Language

Dart will be used to implement application logic, manage data, handle user interactions, and develop reusable application components.

C. Material Design Widgets

Material Design widgets will provide the interface components necessary for creating forms, buttons, navigation menus, dashboard panels, and other application elements.

D. Cupertino Icons

Cupertino Icons will provide additional visual icons that can be used throughout the application's interface.

E. Visual Studio Code

Visual Studio Code will serve as the primary development environment for writing, editing, organizing, and debugging the project's source code.

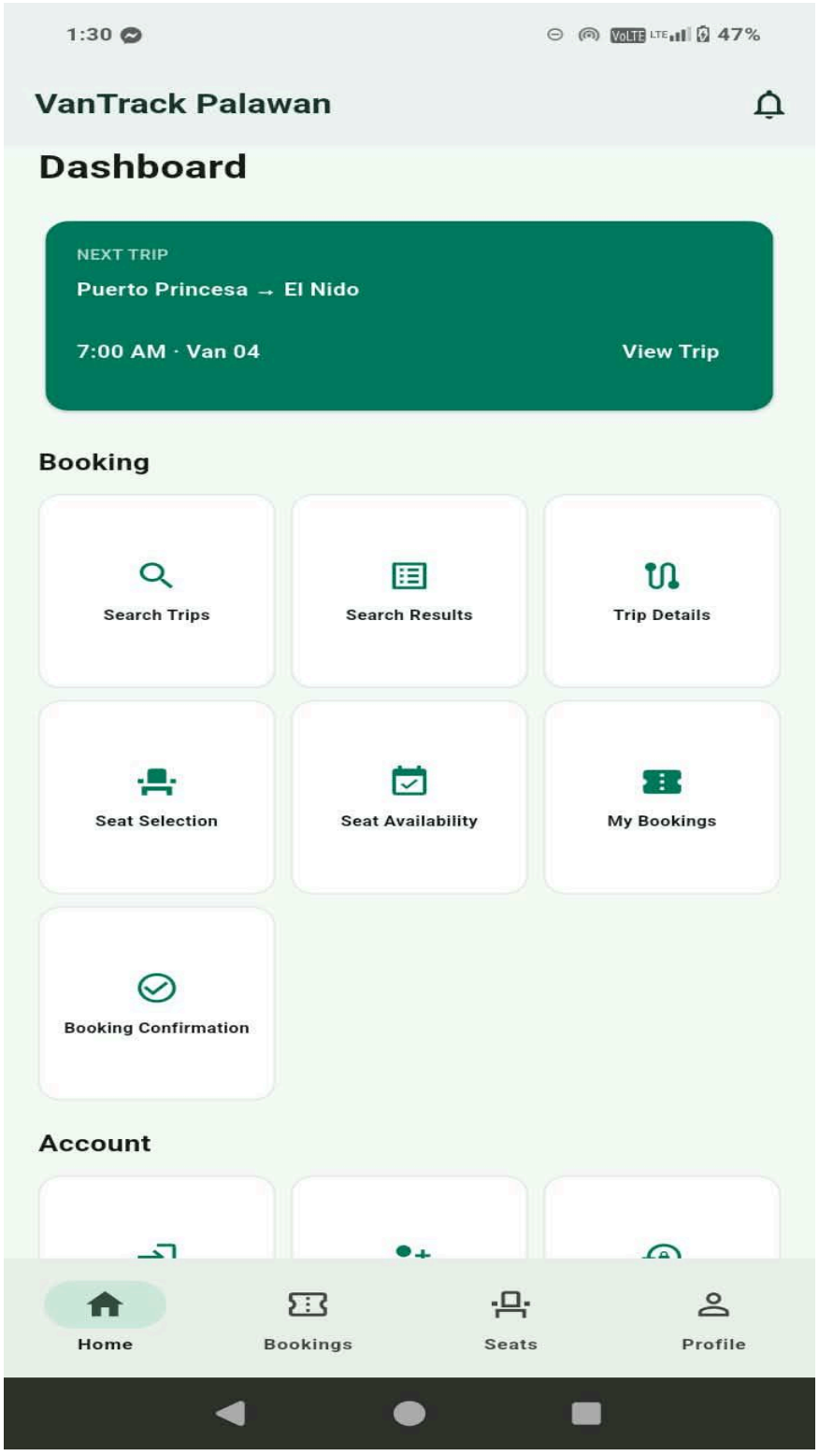
F. Android SDK and Flutter Tools

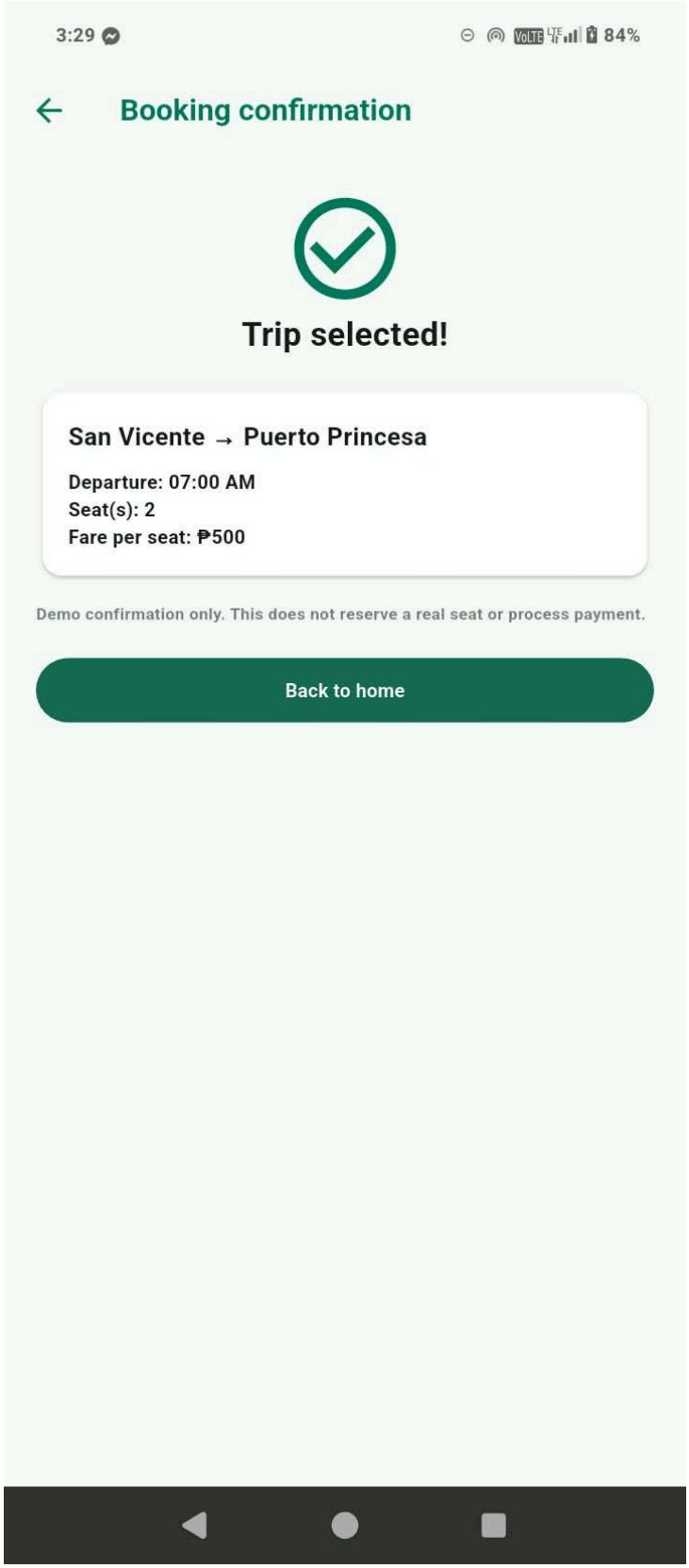
The Android SDK and Flutter development tools will be used to build, run, and test the application on Android devices.

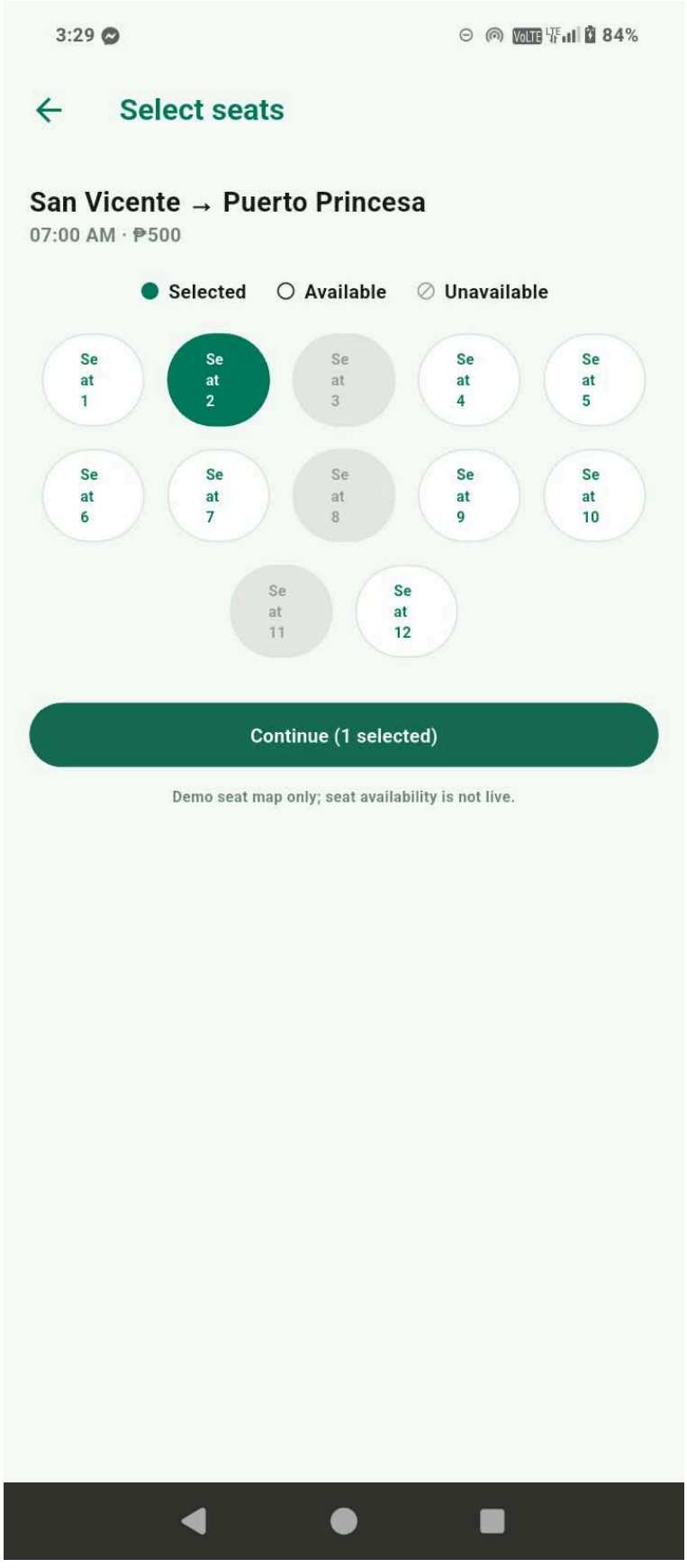
G. Additional Flutter Packages

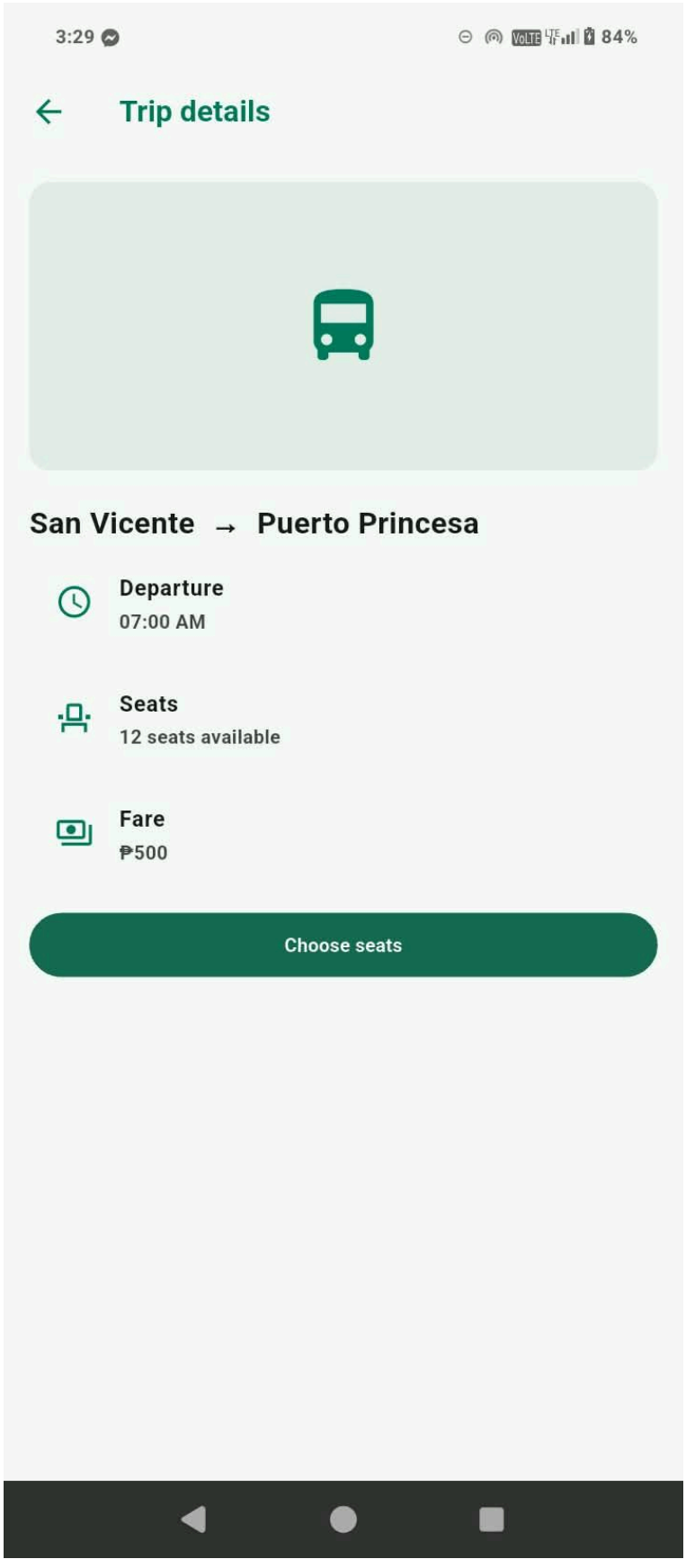
Packages such as shared_preferences and sqflite may be utilized for local data storage and application-related information management. Additional plugins may be integrated when authentication, notifications, and backend connectivity are required.

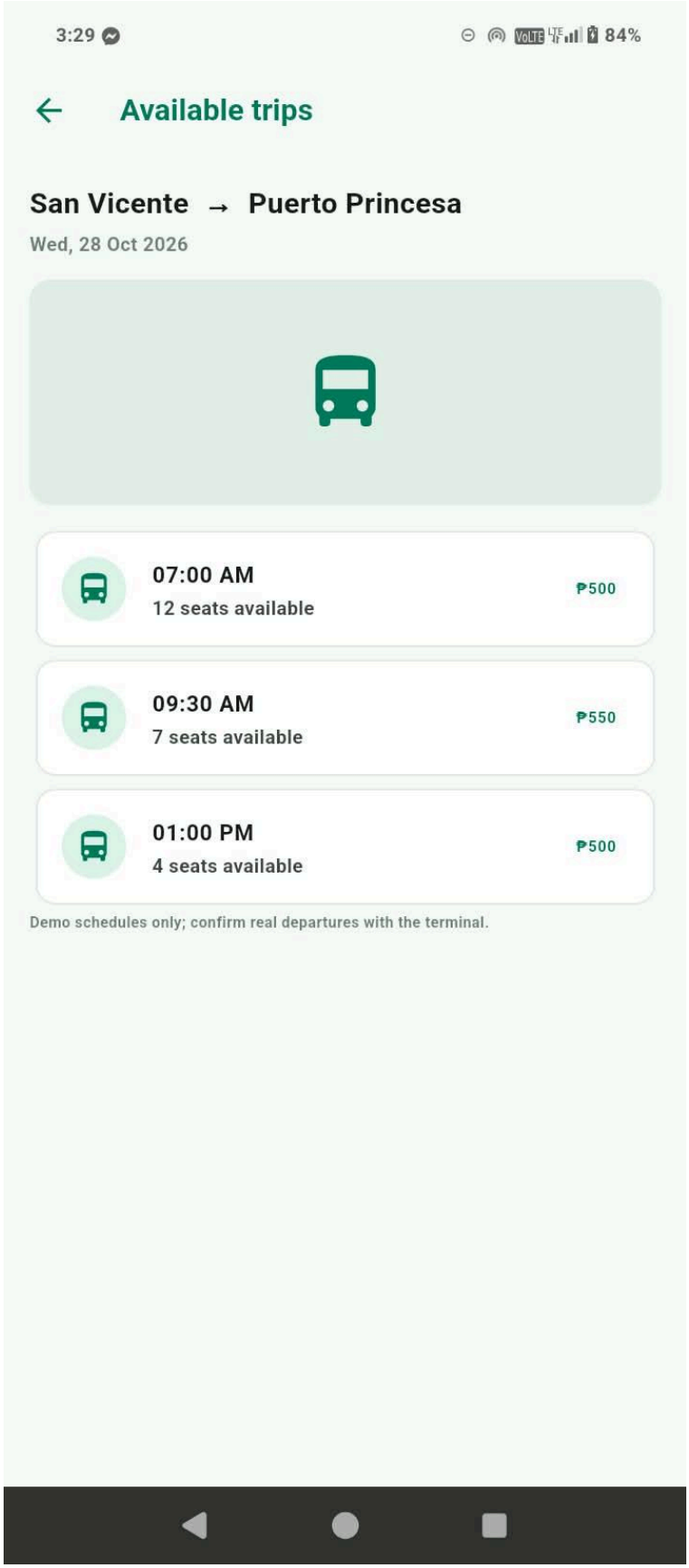
TESTING AND EVALUATION


1:30
47%
VanTrack Palawan
Dashboard
NEXT TRIP
Puerto Princesa → El Nido
7:00 AM · Van 04
View Trip
Booking
Search Trips
Search Results
Trip Details
Seat Selection
Seat Availability
My Bookings
Booking Confirmation
Account
Home
Bookings
Seats
Profile


3:29
84%
Booking confirmation
Trip selected!
San Vicente → Puerto Princesa
Departure: 07:00 AM
Seat(s): 2
Fare per seat: ₱500
Demo confirmation only. This does not reserve a real seat or process payment.
Back to home


3:29
84%
Select seats
San Vicente → Puerto Princesa
07:00 AM · ₱500
Selected
Available
Unavailable
Seat 1
Seat 2
Seat 3
Seat 4
Seat 5
Seat 6
Seat 7
Seat 8
Seat 9
Seat 10
Seat 11
Seat 12
Continue (1 selected)
Demo seat map only; seat availability is not live.


3:29
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Trip details
San Vicente → Puerto Princesa
Departure
07:00 AM
Seats
12 seats available
Fare
₱500
Choose seats


3:29
84%
Available trips
San Vicente → Puerto Princesa
Wed, 28 Oct 2026
07:00 AM
12 seats available
₱500
09:30 AM
7 seats available
₱550
01:00 PM
4 seats available
₱500
Demo schedules only; confirm real departures with the terminal.

Testing and evaluation will be conducted to ensure that VanTrack Palawan functions correctly, provides reliable results, and delivers a user-friendly experience.

The project will utilize the testing frameworks and tools available within the Flutter ecosystem, particularly the Flutter Test framework. These tools will help identify errors, verify application functionality, and evaluate the reliability of individual components and complete application workflows.

The following testing methods will be applied:

A. Unit Testing

Unit testing will be performed to verify individual functions and application logic independently.

The testing process will cover:

- Passenger form validation
- Trip search conditions
- Seat availability calculations
- Rental cost calculations
- Inventory quantity updates

B. Widget Testing

Widget testing will be used to verify that individual user interface components display and behave correctly.

The testing process will cover:

- Login and registration forms
- Dropdown selection
- Dashboard navigation components
- Booking and rental forms
- Buttons and confirmation messages

C. Integration Testing

Integration testing will verify that different application components work together correctly when performing complete user activities.

The testing process will cover:

- Passenger login and trip searching
- Selecting a trip and booking a seat
- Submitting a van rental request
- Recording booking and rental activities
- Updating inventory and reviewing reports

D. Manual Testing and Evaluation

Manual testing will be performed using Google Chrome during web development and an Android device for mobile application testing.

The evaluation will focus on:

- Application navigation and routing
- Screen responsiveness
- User interface consistency
- Form validation
- Button functionality
- Accuracy of displayed information
- Booking and rental workflows
- Overall usability and reliability

Flutter's support for automated testing and testing libraries will help simplify debugging, identify potential issues, and reduce repeated manual testing. The results of these evaluations will be used to improve the application's functionality and overall performance.

PROJECT TIMELINE AND BUDGET

A. Proposed Project Timeline

The development of VanTrack Palawan will follow a structured timeline to organize the planning, design, implementation, testing, and finalization of the application.

The proposed development schedule is presented below.

Development Phase Activities Duration

Planning and Analysis Identify project requirements, target users, and system functions.

Week 1

UI/UX Design Prepare application layouts, dashboard, and navigation.

Week 2

Core Development Develop account management, dashboard, and trip search.

Week 3

Feature Development Develop booking, seat selection, and van rental functions.

Week 4

Management Modules Develop fleet, inventory, history, and reports.

Week 5

Testing and Evaluation Perform unit, widget, integration, and manual testing.

Week 6

Finalization Fix issues, improve the interface, and prepare documentation.

Week 7

Flutter may help expedite the development timeline through its Hot Reload feature, reusable widgets, and shared codebase for Android and iOS platforms.

These capabilities can reduce duplicated development work and potentially minimize the costs associated with application development and maintenance.

B. Project Budget

Flutter is an open-source framework, which provides an opportunity to reduce initial software development expenses.

Its cross-platform capabilities allow developers to maintain a shared codebase for multiple platforms rather than developing entirely separate applications for Android and iOS.

The proposed project budget will consider the following categories:

Budget Category	Description
Development Software	Flutter SDK and development tools
Computer and Equipment	Computer and Android testing device
Internet Connection	Research, package installation, and development

Database and Hosting	Potential backend services and online data storage
Maintenance	Future application updates, testing, and technical improvements

The actual project expenses will depend on the available equipment, internet costs, hosting requirements, and additional services that may be needed during development.

REFERENCES

The following references are presented using the American Psychological Association (APA) format.

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Flutter. (n.d.). Testing Flutter apps. https://docs.flutter.dev/testing/overview

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Tekartik. (n.d.). sqflite package. pub.dev. https://pub.dev/packages/sqflite

DEVELOPER PROFILE

A. Developer Information

Name: Roweto J. Ultra

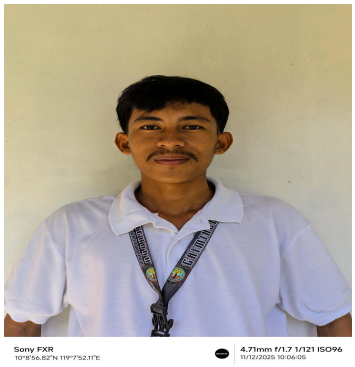
Course and Year Level: BSIT – 3rd Year

Institution:Palawan State University Taytay Campus

Project Title: VanTrack Palawan

Project Role: Flutter Application Developer

B. Developer ID Photo



C. Technical Skills

The developer's technical skills and areas of experience include:

- Flutter Application Development
- Dart Programming
- User Interface Design and Implementation
- Responsive Application Development
- Navigation and Routing

- Form Creation and Validation
- Debugging and Troubleshooting
- Visual Studio Code
- Android Application Testing

D. Project Role and Responsibilities

The developer is responsible for planning, designing, developing, testing, and documenting the VanTrack Palawan application using the Flutter framework.

The responsibilities include implementing the application's user interface, developing its planned features, organizing application navigation, testing user workflows, troubleshooting technical issues, and preparing project documentation.